Week 2 Update

Michael Fang

March 2026

2026-03-18

Week 2 Update

Weekly progress meeting with Professor Torquato.
Main topics:

- Recap of last week
- New patterns analyzed (Classes II, III)
- The Bombieri-Taylor result
- Updated ranking

Week 2 Update
Michael Fang
March 2026

Quick Recap: Last Week

Last week:

- Variance code validated (Poisson benchmark, 1.1% error)
- Three metallic-mean chains: Fibonacci, Silver, Bronze
- All gave $\alpha = 3$ as expected

Pattern	α (measured)	α (expected)	$\bar{\Lambda}$
Fibonacci	3.05	3	0.200
Silver	2.99	3	0.250
Bronze	2.99	3	0.282

This week: Cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.

Quick Recap: Last Week

Last week:

- Variance code validated (Poisson benchmark, 1.1% error)
- Three metallic-mean chains: Fibonacci, Silver, Bronze
- All gave $\alpha = 3$ as expected

Pattern	α (measured)	α (expected)	$\bar{\Lambda}$
Fibonacci	3.05	3	0.200
Silver	2.99	3	0.250
Bronze	2.99	3	0.282

This week: Cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.

Quick reminder of where we left off.
The metallic means all give $\alpha = 3$ because of the 2x2 matrix constraint: $|\lambda_2| = 1/|\lambda_1|$.
Goal this week was to expand beyond just $\alpha = 3$.

Class II: Period-Doubling

The Period-Doubling chain:

$$S \to L, \quad L \to LSS \qquad \Rightarrow \qquad M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$

Matrix gives $\lambda_1 = 2$, $|\lambda_2| = 1$

So: $\alpha = 1 - \dfrac{2\ln(1)}{\ln(2)} = 1$

Results:

- Variance grows like $\ln R$ (not bounded, not power-law)
- $\bar{\Lambda}$ diverges — can't use it to rank this pattern
- This is the boundary case: $\alpha = 1$ exactly
- Class II confirmed

Class II: Period-Doubling
The Period-Doubling chain:
$S \to L, \quad L \to LSS \qquad \Rightarrow \qquad M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$
Matrix gives $\lambda_1 = 2$, $|\lambda_2| = 1$
So: $\alpha = 1 - \dfrac{2\ln(1)}{\ln(2)} = 1$
Results:
- Variance grows like $\ln R$ (not bounded, not power-law)
- $\bar{\Lambda}$ diverges — can't use it to rank this pattern
- This is the boundary case: $\alpha = 1$ exactly
- Class II confirmed

Period-Doubling is the boundary case: $\alpha = 1$ exactly.
$|\lambda_2| = 1$ means $\ln(1) = 0$ in the formula.
First Class II example — variance grows logarithmically.

Class III: 0222 Chain

The 0222 chain:

$$S \to LL, \quad L \to SSLL$$

Here $|\lambda_2| = \sqrt{5} - 1 > 1$
So: $\alpha \approx 0.64$

Results:

- Variance grows like $R^{0.36}$
- $\bar{\Lambda}$ diverges
- Class III confirmed

Now have examples of all three classes.

The three classes:

Class	α	$\sigma^2(R)$
I	> 1	bounded
II	$= 1$	$\sim \ln R$
III	< 1	$\sim R^{1-\alpha}$

The exponent $1 - 0.64 = 0.36$ matches the measured value.

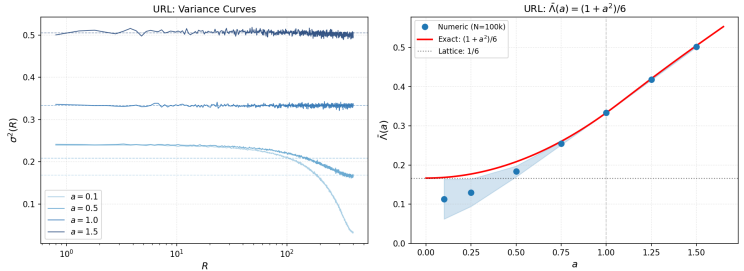
Class III: 0222 Chain
The 0222 chain:
$S \to LL, \quad L \to SSLL$
Here $|\lambda_2| = \sqrt{5} - 1 > 1$
So: $\alpha \approx 0.64$
Results:
- Variance grows like $R^{0.36}$
- $\bar{\Lambda}$ diverges
- Class III confirmed
Now have examples of all three classes.
The three classes:
Class | α | $\sigma^2(R)$
I | > 1 | bounded
II | $= 1$ | $\sim \ln R$
III | < 1 | $\sim R^{1-\alpha}$
The exponent $1 - 0.64 = 0.36$ matches the measured value.

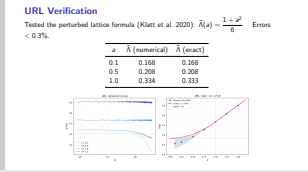
0222 has $|\lambda_2| > 1$, which gives $\alpha < 1$.
The power-law growth $R^{0.36}$ matches the prediction $R^{1-\alpha}$.
All three classes now covered.

URL Verification

Tested the perturbed lattice formula (Klatt et al. 2020): $\bar{\Lambda}(a) = \dfrac{1+a^2}{6}$ Errors $< 0.3\%$.

a	$\bar{\Lambda}$ (numerical)	$\bar{\Lambda}$ (exact)
0.1	0.168	0.168
0.5	0.208	0.208
1.0	0.334	0.333





URL = uncorrelated random lattice (independent displacements on integer lattice).
Exact formula from Klatt et al. (2020): $\bar{\Lambda}(a) = (1+a^2)/6$. Recovers lattice value 1/6 at $a = 0$; equals 1/3 at $a = 1$ (cloaking threshold).
Good independent validation because we have an analytical result for all a. Always $\alpha = 2$ regardless of displacement amplitude a.

Stealthy Patterns

Analyzed stealthy configurations provided by Sam.

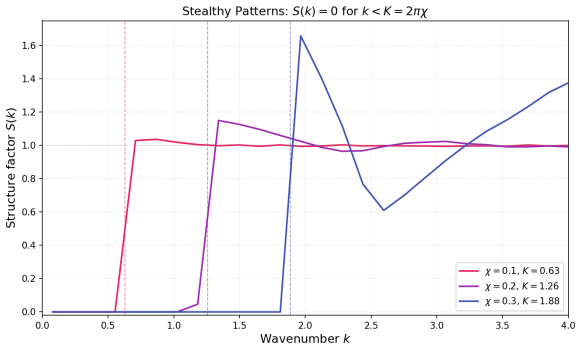
Idea: Force $S(k) = 0$ for all $|k| < K^*$, not just at $k = 0$.

Results ($N = 2{,}000$, $\sim$4000 configs each):

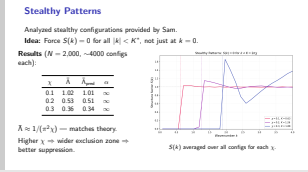
χ	$\bar{\Lambda}$	$\bar{\Lambda}_{\text{pred}}$	α
0.1	1.02	1.01	∞
0.2	0.53	0.51	∞
0.3	0.36	0.34	∞

$\bar{\Lambda} \approx 1/(\pi^2\chi)$ — matches theory.

Higher $\chi \Rightarrow$ wider exclusion zone $\Rightarrow$ better suppression.



$S(k)$ averaged over all configs for each χ.



Data from another grad student (CCO format, 4000 configs per χ).
χ controls the size of the exclusion region: $K = 2\pi\chi$.
Theory predicts $\bar{\Lambda} \approx 1/(\pi^2\chi)$.
The overlay shows all three χ values on one plot — the exclusion zone widens visibly as χ increases, and $S(k)$ rises sharply at K^* before approaching 1 at large k.
$\alpha \to \infty$: $S(k) = 0$ over a finite exclusion zone, not a power law. Exponential spreadability decay confirms this.

Filling the $1 < \alpha < 2$ Gap

There was a gap in the α coverage:

α	Class	Examples
∞	I	Lattice, Stealthy
3	I	Fibonacci, Silver, Bronze
2	I	URL
$1 < \alpha < 2$	I	No quasicrystal
1	II	Period-Doubling
< 1	III	0222

The issue: all 2×2 metallic-mean matrices give $\alpha = 3$ because $|\lambda_2| = 1/|\lambda_1|$.
Solution: the Bombieri-Taylor cubic substitution (1986 paper, p. C3-21).

Filling the $1 < \alpha < 2$ Gap
There was a gap in the α coverage:
The issue: all 2×2 metallic-mean matrices give $\alpha = 3$ because $|\lambda_2| = 1/|\lambda_1|$.
Solution: the Bombieri-Taylor cubic substitution (1986 paper, p. C3-21).

Lattice and stealthy both have $\alpha = \infty$ (S(k) = 0 over a finite region, not just at k=0).
The metallic-mean 2x2 constraint: $\det(M) = \pm1$ forces $|\lambda_2| = 1/|\lambda_1|$, always giving $\alpha = 3$.
The gap at $1 < \alpha < 2$ is filled by the Bombieri-Taylor cubic substitution on the next slide.

Bombieri-Taylor Cubic Substitution

From the 1986 paper (p. C3-21):

Three-letter alphabet:

$$a \to aac,\ b \to ac,\ c \to b$$

Matrix:

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Eigenvalues: $\theta_1 = 2.247$, $|\lambda_2| = 0.802$

Predicted:

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}$$

Implementation:

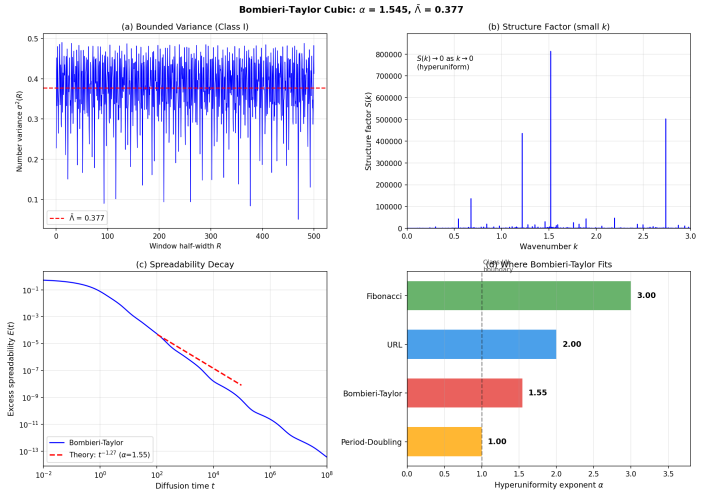
- Extended code for 3-letter alphabets
- Generated 9.5 million points
- Computed variance and $\bar{\Lambda}$

Result: $\bar{\Lambda} = 0.377$, Class I confirmed.

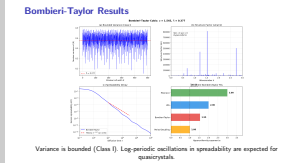
Bombieri-Taylor Cubic Substitution
From the 1986 paper (p. C3-21):
Three-letter alphabet:
$a \to aac,\ b \to ac,\ c \to b$
Matrix:
$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$
Eigenvalues: $\theta_1 = 2.247$, $|\lambda_2| = 0.802$
Predicted:
$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = 1.545$
Implementation:
Extended code for 3-letter alphabets
Generated 9.5 million points
Computed variance and $\bar{\Lambda}$
Result: $\bar{\Lambda} = 0.377$, Class I confirmed.

Key: $|\lambda_2| = 0.802 \neq 1/\theta_1 = 0.445$.
The 3x3 matrix breaks the constraint.
θ_1 is a Pisot number (conjugates all < 1).

Bombieri-Taylor Results



Variance is bounded (Class I). Log-periodic oscillations in spreadability are expected for quasicrystals.



Panel (a): Bounded variance confirms Class I. Oscillates between 0.05–0.49, mean = 0.377.
Panel (b): Structure factor at small k. Shows $S(k) \to 0$ as $k \to 0$ (hyperuniform). Bragg peaks are characteristic of quasicrystals.
Panel (c): Spreadability decay. The log-periodic oscillations make power-law fitting unreliable for quasicrystals. The eigenvalue formula is exact; spreadability fit is not.
Panel (d): Shows Bombieri-Taylor fills the gap between Period-Doubling ($\alpha = 1$) and URL ($\alpha = 2$). Stealthy patterns have $\alpha \to \infty$ and do not appear on this axis.
Confidence: $\alpha = 1.545$ is exact (analytical from eigenvalues). $\bar{\Lambda} = 0.377$ is numerical average of bounded variance.

Updated Ranking

Here's where everything stands now:

Pattern	Class	α	$\bar{\Lambda}$
Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
Silver	I	3.0	0.250
Bronze	I	3.0	0.282
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	∞	0.358
Bombieri-Taylor	**I**	**1.545**	**0.377**
Period-Doubling	II	1.0	diverges
0222	III	0.64	diverges

Updated Ranking
Here's where everything stands now:

Pattern	Class	α	$\bar{\Lambda}$
Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
Silver	I	3.0	0.250
Bronze	I	3.0	0.282
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	∞	0.358
Bombieri-Taylor	**I**	**1.545**	**0.377**
Period-Doubling	II	1.0	diverges
0222	III	0.64	diverges

26 patterns total (counting all URL/Gaussian/Stable/Stealthy variants). Table shows a selected subset sorted by $\bar{\Lambda}$ within each class.

Copper ($\bar{\Lambda} = 0.293$) and Nickel ($\bar{\Lambda} = 0.310$) fill the gap between Bronze (0.282) and URL $a = 1.0$ (0.333) in the metallic-mean family.

Bombieri-Taylor fills the gap between $\alpha = 1$ and $\alpha = 2$.

First quasicrystal with $1 < \alpha < 2$.

Summary

This week:

- Added Class II (Period-Doubling, $\alpha = 1$)
- Added Class III (0222, $\alpha = 0.64$)
- Verified URL formula to $< 0.3\%$
- Analyzed stealthy data ($\bar{\Lambda} \approx 1/(\pi^2\chi)$)
- Implemented Bombieri-Taylor cubic ($\alpha = 1.545$)

Main result: Filled the $1 < \alpha < 2$ gap with a quasicrystal.

Questions:

- Are there other cubic/quartic substitutions to try?
- Should 2D generalizations be next?

Summary

This week:
- Added Class II (Period-Doubling, $\alpha = 1$)
- Added Class III (0222, $\alpha = 0.64$)
- Verified URL formula to $< 0.3\%$
- Analyzed stealthy data ($\bar{\Lambda} \approx 1/(\pi^2\chi)$)
- Implemented Bombieri-Taylor cubic ($\alpha = 1.545$)

Main result: Filled the $1 < \alpha < 2$ gap with a quasicrystal.

Questions:
- Are there other cubic/quartic substitutions to try?
- Should 2D generalizations be next?

Total patterns analyzed: 26
Key accomplishment: Bombieri-Taylor fills the gap.
Possible next steps: other cubic substitutions, 2D/3D.

References

- Torquato & Stillinger (2003), Phys. Rev. E 68, 041113
- Bombieri & Taylor (1986), J. Physique Colloques 47, C3-19
- Oğuz et al. (2017), Phys. Rev. B 95, 054119
- Klatt, Kim & Torquato (2020), Phys. Rev. E 101, 032118
- Torquato (2021), Phys. Rev. E 104, 054102

References

- Torquato & Stillinger (2003), Phys. Rev. E 68, 041113
- Bombieri & Taylor (1986), J. Physique Colloques 47, C3-19
- Oğuz et al. (2017), Phys. Rev. B 95, 054119
- Klatt, Kim & Torquato (2020), Phys. Rev. E 101, 032118
- Torquato (2021), Phys. Rev. E 104, 054102

Main references used this week.